

INFO 5602 – Information Visualization

Course Syllabus

Course Dates:

Dates: 2026-08-20 through 2026-12-04

Course Schedule:

Credit Hours: 3

Grading Mode: Letter

Campus: Boulder Main Campus

Location: Boulder Main Campus

Instruction Mode: In Person

Session: Boulder Main Campus Semester

Dates: 2026-08-20 through 2026-12-04

Schedule & Location

TTh 11am-12:15pm [Hellems Arts & Sciences Bldg N270](#)

Instructor:

Abel Iyasele, PhD, MS, MBA

Teaching Assistant Professor

Email: abel.iyasele@colorado.edu

Office Hours:

Wednesdays, 10AM – 11AM

Send email to me (abel.iyasele@colorado.edu) to schedule appointment for videoconference.

Course Description (copied from classes.colorado.edu):

Explores the design, development and evaluation of information visualizations. Covers visual representations of data and provides hands-on experience with using and building exploratory tools and data narratives. Students create visualizations for a variety of domains and applications, working with stakeholders and their data. Covers interactive systems, user-centered and graphic design, perception, data storytelling and analysis, and insight generation. Programming knowledge is strongly encouraged.

The big picture of the course is achieving effective stakeholder communication with information visualization. One motivation for INFO 5602 is that, in the real-world, stakeholders are attention-deficient and poorly produced charts are prevalent; effective stakeholder communication is thus problematic. This course will take a storytelling approach to Information Visualization, which helps focus you on stakeholders, stakeholder communication and management.

Students will learn about decomposing charts and presentations into *signal* and *noise* components and how to improve the *signal* in charts and presentations to achieve effective communication with attention-deficient stakeholders in the real-world. Students will also critique, create and present visualizations to their colleagues in class as part of group learning.

There are at least two approaches to learning information visualization: a tool-first approach and a design/storytelling – first approach. We take a design/storytelling - first approach to this course. This course was structured to intentionally shift your understanding of information visualization from focusing on tools for chart creation to focusing on using data and charts for telling stories with stakeholders in mind.

If your expectation is to focus on tools, or you are unwilling to learn the design/storytelling-first approach, this course may not be for you, and you should reconsider taking this course.

While this class is not about and will not teach any one specific visualization tool, you will get to work on and submit individual projects – using any visualization tool of your choice (e.g. Tableau, PowerBI, Excel, PowerPoint etc.). You will receive web links to learn visualization tools like PowerBI, Tableau, PowerPoint etc. privately on your own for popular professional projects in real-world settings. You will also get to work with your colleagues in a group project using these tools.

Many parts of this course will feel repetitive. Repetition is one method for storytelling – and for retaining information. The goal is to use repetition help you build muscle memory about design concepts using charts – either individually or in a dashboard.

We will aim to cover everything in the syllabus; we may or may be unable to cover everything in this syllabus because of unforeseen/constraining circumstances.

INFO 5602 will also cover how to use Python Seaborn to produce various univariate and multivariate plots for data storytelling, analysis and insight generation. We will be using the Anaconda distribution for Python in class. This course assumes a basic knowledge of Python and that you can access Python Jupyter notebooks to complete assigned homework. I will also expect that you can access and upload the Jupyter notebooks that I will be posting in Canvas for class discussions and your homework. To follow the discussion in class, it will help if you have or can download Anaconda 3 on your laptops – and have it loaded with the respective Jupyter notebook for each class.

Course Learning Outcomes

Upon successful completion of this course, you will be able to:

- understand and describe the socio-cultural challenge of *attention-deficiency* and its negative impact on effective stakeholder communication and decision-making in real-world settings.
- understand the prevalence of bad charts in the real-world, and how to choose the right chart among available options to present information.
- understand the role *context* plays in designing and implementing data visualization.
- identify *noise* and *signal* in charts and presentations. Use pre-attentive attributes to boost chart/presentation signals.
- work individually to design, implement and present a simple visualization project using a tool of your choice.
- work with other students in groups to design and implement simple visualization projects

- understand the differences between exploratory data analysis and *explanatory data analysis*.
- understand user-centered design (UCD)
- use Python Seaborn to visually represent and describe individual numeric variables in a given dataset.
- use Python Seaborn to visually represent and describe the relationship between numeric variables in a given dataset.
- use visualization to gain insights from raw data and tell stories to stakeholders.
- summarize peer-reviewed journal papers and other scholarly materials on Information Visualization

Course structure

• Part 1: (~ 1 – 4 weeks): Class lectures

- Class lectures focused on theoretical aspects of information visualizations. You learn about the problem of stakeholder attention deficiency, context, cognitive load, decomposing presentations into *signal* and *noise*, the tools you can use to identify *noise* and pre-attentive attributes – tools you can use to boost *signal* for effective communications with attention-deficient stakeholders.
- Introduction to the *50-Chart visualization Challenge*
- Learn about storytelling
- Summarize key journal papers focused on information visualization.
- ~3 assignments; Exam 1

• Part 2 (~ 2 – 3 weeks): Insight generation from Python-generated charts

- Work with univariate plots in Python Seaborn to summarize numeric data/variables and draw actionable insights.
- Work with multivariate plots in Python Seaborn to examine relationships between numeric variables and draw actionable insights.
- ~3 assignments; Exam 2

• Part 3 (~ 6 weeks): Focus on group learning/in-class activity

- Students present their individual projects to their colleagues (proxy for real-world attention-deficient audience) in class and receive feedback.
- Students use various tools of their choice (like PowerBI, Tableau, Excel etc) and real-world data to produce and submit dashboards for grading.
- Group project submission/Exam 3
- Individual projects (two) presentation and submission

Required Technology, Course Textbooks and Materials

Technology:

- Python Anaconda 3.x distribution – make sure you have the current version of Anaconda to help you follow classes and do your Homeworks
- Jupyter notebooks – Classes and Homeworks will be in Jupyter notebooks, and PowerPoint presentations. Make sure you know how to upload classes into your Anaconda page to aid your work
- Access to Tableau, Power BI and other visualization tools is encouraged for your individual projects

Required Texts: none

Recommended Texts:

- Ozdemir, Sinan (2024) *Principles of Data Science* Packt Publishing ISBN: 978 – 1 – 83763-630-3
- Vaughan, Daniel (2024) *Data Science: The Hard Parts*. O'Reilly. ISBN: 978-1-098-14647-4
- Gutman, Alex J., Goldmeier Jordan (2021) *Becoming a Data Head: How to Think, Speak, and Understand Data Science, Statistics, and Machine Learning*. Wiley, ISBN 9781119741749
- Knafllic, Cole Nussbaumer (2015) *Storytelling with Data: A Data Visualization Guide for Business Professionals*; Wiley ISBN 9781119002253
- Embarak, Ossama (2018): *Data Analysis and Visualization Using Python*. Apress ISBN 978-1-4842-4109-7
- Raman, Kirthi (2015) *Mastering Python Data Visualization*: Packt Publishing ISBN 978-1-78398-832-7
- Loudermilk, T (2013) *User-Centered Design*. O'Reilly Media Inc ISBN 978-1-419-35980-5

Course Expectations, Requirements, and Policies

Expectations for Course Participation: *You should plan to spend around 12 – 15 hours per week on the course. Assignments will be due approximately weekly.*

Assignments: During the semester I will assign, collect, and grade assignments. There will be approximately seven (7) formal assignments throughout the semester. **All homework/assignments are due 11:59PM Boulder local time as specified below**

You may receive assistance from other students in the class and me, but your submissions must be composed of your own work.

Exams: There will be two (2) exams. Students will also have the opportunity to opt for a third exam in lieu of submitting a group project. **All exams will be online and take-home.**

Individual Projects: Each student will be expected to undertake, present and submit **two** individual projects based on the concepts learned in this course.

Group Project: Each student will be expected to work with their colleagues to undertake and submit a Group Project based on the concepts learned in this course. Students can elect to take a third Exam in lieu of working on and submitting a Group Project.

Grading: You should be able to explain your work on assignments and exams and your rationale. Based on your explanation (or lack thereof), I may modify your grade.

Late Work Policy: Late assignments will not be accepted without prior consent of the instructor. I may impose a penalty for accepting any late assignment submission.

Attendance: I will be tracking attendance, and it will be 15% of your grade. See below Final Grade Calculation section.

Final Grade Calculation

Assessment Type	Number	% of course grade
Assignments/Homework	~Seven (7)	25%
Exams (Exam 1 and Exam 2)	Two (2)	20%
Group Project/Exam 3	One (1)	20%
Individual Project	Two (2)	20%
Physical class attendance**		15%
Total		100%

****Count of how many times you signed the attendance sheet / total number of classes held*100**

Grading Scale

Final Percent	Grade
93%-100%	A
90%-92%	A-
88%-89%	B+
83%-87%	B
80%-82%	B-
78%-79%	C+
73%-77%	C
70%-72%	C-
60%-69%	D
Under 60%	F

Weekly Schedule

Week Number	Calendar Date	Activity/Description	Remark
1	8/17 – 8/21	- Syllabus review and semester expectation setting - The challenge of attention-deficiency, poor visualizations and stakeholder decision-making in the real world - Understanding information visualization as a tool to achieve effective stakeholder communication. - Chapter 0: Learning about the prevalence of bad graphs/charts in the real world - Chapter 1: Context, audience, problem definition and data visualization - Chapter 2: Choosing an effective visual to represent data	- Homework 1 – 3 issued - Homework 1 – 3 due (multiple dates from week 1 to week 4)
2	8/24 – 8/28		
3	8/31 – 9/4		
4	9/7 – 9/11	- Chapter 3: Data visualization, the problem of clutter and the concept of cognitive load - Constructively critiquing data visualizations - Introducing <i>The 50 visualization chart challenge</i> - Decomposing charts into <i>signal</i> and <i>noise</i> for effective stakeholder communication - Chapter 4: Focusing stakeholder attention using <i>pre-attentive attributes</i>	- Homework 4 – 7 issued - Homework 4 – 7 due (multiple dates from week 4 to week 7)
5	9/14 – 9/18		
6	9/21 – 9/25	September 24: Exam 1 (No class on Sep. 24)	
7	9/28 – 10/2	- Introduction to User-Centered design and how to think like a designer - Storytelling and information visualization Focus on univariate descriptive statistics and insight generation using Python Seaborn	

		• Kernel Density Estimate KDE plots and Histograms/Distribution plots • Box plot • Violin plot • Swarmplot Focus on visualizing statistical relationships and insight generation using Python Seaborn • Scatter plot • Regplot • Pairplot • Facet Grid Watch and discuss dashboard presentation videos by experts from Tableau, PowerBI etc.	
8	10/5 – 10/9	• Individual Project Presentations (Part 1): Real World Chart Evaluations	
9	10/12 – 10/16		
10	10/19 – 10/23		
11	10/26 – 10/30	• October 29: Exam 2 (No class on Oct. 29)	
12	11/2 – 11/6	• Individual Project Presentations (Part 2): Dashboards	
13	11/9 – 11/13		
14	11/16 – 11/20		
15	11/23 – 11/27	• Fall Break (Nov. 23 – Nov. 27) • No Classes	
16	12/1 – 12/5	• Dec. 1: Final Day of class for INFO 1301 • Dec. 4 - 5: Exam 3 (No class on Dec. 5) • Dec. 5: Group Project submission	
17+	Dec. 7	• Fall 2026 Deadline for Instructors to Post Grades (11:59 p.m.)	

Syllabus Statements

Honor Code

All students enrolled in a University of Colorado Boulder course are responsible for knowing and adhering to the [Honor Code](#). Violations of the Honor Code may include but are not limited to: plagiarism (including use of paper writing services or technology [such as essay bots]), cheating, fabrication, lying, bribery, threat, unauthorized access to academic materials, clicker fraud, submitting the same or similar work in more than one course without permission from all course instructors involved, and aiding academic dishonesty. Understanding the course's syllabus is a vital part in adhering to the Honor Code.

All incidents of academic misconduct will be reported to Student Conduct & Conflict Resolution: StudentConduct@colorado.edu. Students found responsible for violating the [Honor Code](#) will be assigned resolution outcomes from the Student Conduct & Conflict Resolution as well as be subject to academic sanctions from the faculty member. Visit [Honor Code](#) for more information on the academic integrity policy.

Accommodation for Disabilities, Temporary Medical Conditions, and Medical Isolation

If you qualify for accommodations because of a disability, please submit your accommodation letter from Disability Services to your faculty member in a timely manner so that your needs can be addressed. Disability Services determines accommodations based on documented disabilities in the academic environment. Information on requesting accommodations is located on the [Disability Services website](#). Contact Disability Services at 303-492-8671 or DSinfo@colorado.edu for further assistance. If you have a temporary medical condition, see [Temporary Medical Conditions](#) on the Disability Services website.

If you have a temporary illness, injury or required medical isolation for which you require adjustment, you can send me an email to notify me about absence due to illness, injury, or medical isolation. Because of FERPA student privacy laws, I will not require students to state the nature of their illness when alerting me. Furthermore, I cannot require "doctor's notes" for classes missed due to illness – so do not send them to me; campus health services no longer provide "doctor's notes" or appointment verifications. Also see [Temporary Medical Conditions](#) on the Disability Services website.

Accommodation for Religious Obligations

Campus policy requires faculty to provide reasonable accommodations for students who, because of religious obligations, have conflicts with scheduled exams, assignments or required attendance. Please communicate the need for a religious accommodation in a timely manner. In this class, let me know if any key dates e.g. dates for exams do not allow you complete these activities.

[See the Student Academic Accommodations for Religious Observances Policy](#) for more information

Preferred Student Names and Pronouns

CU Boulder recognizes that students' legal information doesn't always align with how they identify. Students may update their preferred names and pronouns via the student portal; those preferred names and pronouns are listed on instructors' class rosters. In the absence of such updates, the name that appears on the class roster is the student's legal name.

If you wish to have a name other than your legal name appear on your instructors' class rosters and in Canvas, or if you wish to choose pronouns to appear on your instructors' class rosters and in Canvas, visit the [Registrar's website](#) for instructions on how to change your personal information in university systems

Classroom Behavior

Students and faculty are responsible for maintaining an appropriate learning environment in all instructional settings, whether in person, remote, or online. Failure to adhere to such behavioral standards may be subject to discipline. Professional courtesy and sensitivity are especially important with respect to individuals and topics dealing with race, color, national origin, sex, pregnancy, age, disability, creed, religion, sexual orientation, gender identity, gender expression, veteran status, marital status, political affiliation, or political philosophy.

Additional classroom behavior information

- [Student Classroom and Course-Related Behavior Policy](#).
- [Student Code of Conduct](#).

- [Office of Institutional Equity and Compliance](#).

Sexual Misconduct, Discrimination, Harassment and/or Related Retaliation

CU Boulder is committed to fostering an inclusive and welcoming learning, working, and living environment. University policy prohibits [protected-class](#) discrimination and harassment, sexual misconduct (harassment, exploitation, and assault), intimate partner abuse (dating or domestic violence), stalking, and related retaliation by or against members of our community on- and off-campus. The [Office of Institutional Equity and Compliance \(OIEC\)](#) addresses these concerns, and individuals who have been subjected to misconduct can contact OIEC at 303-492-2127 or email CUreport@colorado.edu. Information about university policies, [reporting options](#), and [OIEC support resources](#) including confidential services can be found on the [OIEC website](#).

Please know that faculty and graduate instructors are required to inform OIEC when they are made aware of incidents related to these concerns regardless of when or where something occurred. This is to ensure that individuals impacted receive outreach from OIEC about their options and support resources. To learn more about reporting and support for a variety of concerns, visit the [Don't Ignore It page](#).

Mental Health and Wellness

The University of Colorado Boulder is committed to the well-being of all students. If you are struggling with personal stressors, mental health or substance use concerns that are impacting academic or daily life, please contact [Counseling and Psychiatric Services \(CAPS\)](#) located in C4C or call (303) 492-2277, 24/7.

Free and unlimited telehealth is also available through [Academic Live Care](#). The Academic Live Care site also provides information about additional wellness services on campus that are available to students.

Acceptable Use of AI (artificial intelligence) in this Class

You may use technology tools like Grammarly to fix grammatical errors in your assignments, but you may not use artificial intelligence tools like ChatGPT, Gemini etc to create, copy, enhance or otherwise embellish answers to questions in your assignments, exams or projects. This course will use a tool like *Turnitin* to look for plagiarism cases for further action.